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pressures and the mole fractions of two components A & B, which is known as Raoult's Law. This law states that "in the case of a solution of volatile liquids, the partial vapour pressure of each component (A & B) of the solution is directly proportional to its mole fraction".

According to Raoult's law,

$$p_A \alpha x_A \qquad (9.3)$$

$$p_A = k\, x_A$$

when $x_A = 1$, $k = p_A^\circ$

where p_A° is the vapour pressure of pure component 'A' at the same temperature.

Therefore,

$$p_A = p_A^\circ x_A \qquad (9.4)$$

Similarly, for component 'B'

$$p_B = p^\circ_B x_B \qquad (9.5)$$

x_A and x_B are the mole fraction of the components A and B respectively.

According to Dalton's law of partial pressure the total pressure in a closed vessel will be equal to the sum of the partial pressures of the individual components.

Hence,

$$P_{total} = p_A + p_B \qquad (9.6)$$

Substituting the values of p_A and p_B from equations (9.4) and (9.5) in the above equation,

$$P_{total} = x_A p_A^\circ + x_B p_B^\circ \qquad (9.7)$$

We know that $x_A + x_B = 1$ or $x_A = 1 - x_B$

Therefore,

$$P_{total} = (1 - x_B)\, p_A^\circ + x_B\, p_B^\circ \qquad (9.8)$$

$$P_{total} = p_A^\circ + x_B (p_B^\circ - p_A^\circ) \qquad (9.9)$$

The above equation is of the straight-line equation form y = mx+c. The plot of P_{total} versus x_B will give a straight line with $(p_B^\circ - p_A^\circ)$ as slope and p_A° as the y intercept.

Let us consider the liquid solution containing toluene (solute) in benzene (solvent).